Andrew Silvestri

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Chemical engineering student with interests in energy systems, process simulation, and energy modeling

EDUCATION

The University of Texas at Austin, Cockrell School of Engineering **Expected May 2028**

Bachelor of Science, Chemical Engineering | Minor, Energy Studies | GPA: 3.06/4.00

Relevant Coursework: Physical Chemistry, Organic Chemistry I & II, Electricity Systems, Energy Law & Policy, Energy Technology & Policy. In progress: Transport Phenomena, Thermodynamics, Numerical Methods in ChE, Applied Statistics.

WORK EXPERIENCE

Process Engineering Intern, Targa Resources **May 2026 – Aug. 2026**

- Process and pipeline engineering for natural gas gathering systems; process simulation in ProMax.

Owner, Silver Tree Property Services LLC **Aug. 2023 – Present**

- Founded and run a property services business; price and scope jobs, source materials, and manage clients.
- Tracked costs and margins in Excel for properties and projects.

Lab Technician (Internship), Industrial Oils Unlimited **May 2023**

- Tested in-house industrial lubricant blends, currently in research trials, using viscosity, FTIR, and metal content analysis.
- Diagnosed solution defects through IR spectroscopy and reported findings in Excel.

Camera/Sound Technician, St. John's Episcopal Church **Oct. 2021 – Jul. 2023**

- Operated multiple cameras with DaVinci Resolve and ATEM switchers; mixed livestream audio on a Yamaha console.

PROJECTS

Energy Systems Model, Personal Research Project **2026 – Present**

- Simulates a supply shock through plant capacity, fuel prices, and demand; checked against 11 events since 1973.
- 2 TB dataset from 83 sources (EIA, ENTSO-E, ERA5, EM-DAT) pulled in Python; monthly country panel from 15 of them.

andrewsilvestri.com, Interactive Energy Tools **2026 – Present**

- Grid-battery revenue simulator: linear program over 8,760 hourly prices; value flattens above four hours of duration.
- Industrial heat electrification break-even (electric vs. gas boiler at Texas prices); life-cycle climate cost of 20 products.

INVOLVEMENT

Kay Bailey Hutchison Energy Center, Energy Studies Cohort **Jun. 2025 – Aug. 2025**

Texas NET Project, DAC Chemical Team Member **Jan. 2025 – Present**

Kay Bailey Hutchison Energy Center, Student Advisory Council **Jan. 2025 – Present**

SKILLS

- **Technical Experience:** ProMax, Python (pandas, scikit-learn, Manim), Julia, MATLAB, JMP, Excel (advanced, VBA), Altair HyperStudy, LaTeX, Zotero
- **Lab Experience:** Viscosity (capillary viscometer), FTIR, specific gravity, metals, freeze point, gel electrophoresis

HONORS & AWARDS

LeaderShape Institute, Engineering Leadership Retreat **2025**

STEM Muse Mentor Program, Accepted Mentee **2025**

James W. and Glena C. Kilmer Engineering Scholarship Recipient **2023**

National Merit Commended Scholar **2023**

Eagle Scout Rank Award & Outstanding Eagle Scout Leadership Service Project Award **2021**

National Advanced Youth Leadership Experience, Participant (2020), Staff (2021, 2022)

5x Marathoner with goal to run Earth's circumference (24,901 miles)